

Sun, Light & the Skin

Melanin, vitamin D, and the single habit that protects more than any product you will ever sell

Learning Objectives

- Explain how UVA and UVB radiation each affect the skin differently
- Describe how vitamin D synthesis works, and the genuine trade-off with sun protection
- Explain melanin's protective role and why sun risk still exists across every skin tone
- Teach sunscreen as the non-negotiable final step of every daytime routine, and mean it

The most important lesson in this course

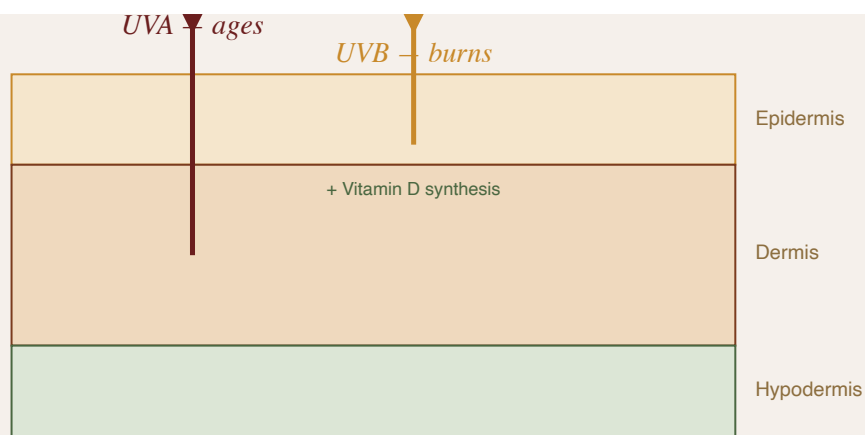
If a client remembers only one thing you ever teach her, let it be this module. No serum, no facial, no ritual does more for long-term skin health than consistent sun protection.

Every other step in a routine is optional on a given day. This one never is.

UVA and UVB — two different threats

- **UVA rays** — longer wavelength, penetrating deep into the dermis. Present all day, all year, even through glass. Responsible for premature ageing.
- **UVB rays** — shorter wavelength, acting mainly on the epidermis. The primary cause of sunburn — and the trigger for vitamin D synthesis.

Remember it simply: **UVA ages, UVB burns** — a protective sunscreen must guard against both, described on a label as "broad-spectrum."



UVA penetrates to the dermis and drives ageing; UVB acts on the epidermis, causing burn and triggering vitamin D synthesis.

The vitamin D story — and the honest trade-off

Consider a professional athlete, diligent about sunscreen, whose vitamin D reads surprisingly low at his check-up. The reason: UVB striking the skin converts a precursor into pre-vitamin D3, and sunscreen, working as intended, blocks much of that UVB. This is not a flaw in sunscreen — it is an honest trade-off worth naming to clients. The responsible answer is diet, supplementation where advised, and brief deliberate exposure — not daily unprotected midday sun.

Melanin — real protection, not immunity

Melanin genuinely absorbs and scatters UV radiation, and deeper skin tones carry a measurably higher natural SPF equivalent. This is real protection — and also, dangerously, the reason sun damage is so often under-diagnosed on deeper skin, because both clients and practitioners wrongly assume it "doesn't need" sunscreen. It does.



The Origin Bubble Aboriginal Australia · Ochre · Deep time

The Earth Worn as a Second Skin

Across Aboriginal Australia, ochre — iron-rich earth pigment ground from stone — has been worn on the skin for tens of thousands of years, among the longest continuously practised skin traditions known anywhere on earth. Ground fine and mixed with fat or water, it served ceremony, identity, and story.

It also, observably, served practicality: applied across skin already deeply protected by generations of melanin, ochre's mineral density offered a further physical barrier against the harshest, most relentless sun on the planet.

No elder needed a spectrometer to know this — the knowledge was observed and passed on, unbroken, for immeasurable time.

The science

Iron-oxide pigments like ochre have measurable UV-attenuating properties — a genuine physical barrier layered atop deeply melanated skin's substantial natural protection.

Why this belongs here: No skin, anywhere, has ever truly been "sun-proof" — worth remembering when a client assumes her deep tone means she can skip this step.

The non-negotiable

A broad-spectrum sunscreen, SPF 30 or higher, applied every single morning regardless of weather or tone, reapplied through real sun exposure, is the single highest-leverage action for skin's long-term health. It never becomes optional.

Key Terms

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[object Object]

1

[object Object]

2

[object Object]

3

[object Object]

Practical Activity · The Sun Conversation

Write out how you would explain sun protection to two clients: one who believes sunscreen is unnecessary for her deep tone, and one worried about vitamin D

deficiency. **Practise saying it aloud — you will give this conversation for the rest of your career.**